INSTRUMENT DOCUMENTATION · USER'S GUIDE

Status Panel — User's Guide

The five-minute version: what the wall is telling you, which buttons do what, how to answer an alarm, and how to shape the rotation from the dashboard. **State note (21 October 2026):** this guide describes the panel contract and a recorded deployment topology; the live daemon host, listener, and certificate state have not been rechecked for this edition. For the full mechanism, colour law, and per-pixel anatomy, read the Operator Manual.

What you are looking at

A Pimoroni Galactic Unicorn — 53×11 LEDs on the wall — rendering live SolariNet fleet health. **xenon** computes the state and serves it; the **solariPanel** daemon on **lithium** polls it every 5 s and frames it to the panel over USB every 2 s; the panel firmware animates at 25 Hz from the last snapshot it received. The panel never computes fleet state — it renders what arrived, and it tells you loudly when nothing is arriving.

THE THREE IDEAS THAT EXPLAIN EVERYTHING

Twelve screens	Four themes — **A Abstract**, **B Text**, **C Charts**, **D Instruments** — of three screens each, rotating on a dwell you control. The panel boots on D1, the ambient pool view.
Colour is state	Green healthy · azure quiet/informational · amber degraded or busy · red down or critical · violet maintenance · slate unknown. Brightness carries magnitude or tier, hue carries condition — everywhere, on every screen.
One alarm	When a critical condition arms, an inlay and a top-right beacon appear, a two-tone sounds, and *every button becomes an acknowledge*. Acknowledging silences that episode; a new episode re-arms.

WHERE THE CONTROLS LIVE

Nine buttons on the panel itself (next page), and the **Panel** page of the dashboard at `dashboard.akoria.net` — theme, screen, brightness, dwell, per-screen rotation and weights, sleep and wake. A setting is only *confirmed* when the panel itself reports it back; the dashboard shows pending until then.

> **If the panel looks wrong, first decide: fleet fault or panel fault?** A red screen full of DOWN is the fleet. `NO DATA` or `LINK LOST` printed on the matrix is the panel's own data path — see the last page.

ON THE PANEL

Buttons

Four theme keys, two brightness, two sound-and-help, one sleep — and one overriding rule when the alarm is armed.

While an alarm is armed, every button is an acknowledge. Press A expecting a theme change and you will silence the alarm instead. Ack first, then press again for what you wanted.

BUTTON	NORMAL ACTION	DETAIL
A	Theme ABSTRACT	Jumps to the theme's first *enabled* screen; rotation continues from there.
B	Theme TEXT	As above.
C	Theme CHARTS	As above.
D	Theme INSTRUMENTS	As above. D1 is where the panel boots.
LUX +	Brightness +5%	Latches manual — the light sensor stops driving brightness until AUTO is re-enabled from the dashboard's Panel page.
LUX −	Brightness −5%	Same latch. The driver floors at 25%, so the panel dims down but never out.
VOL +	Help overlay	Two lines of help for the screen currently showing. Press VOL+ again to close, or wait 30 s. Any other button closes it *and* performs its normal action. Rotation pauses while help is open. Ignored while an alarm is armed — the ack wins.
VOL −	Alarm tone quieter	Steps of 0.1, and it is the *only* volume control — there is no louder button. A reboot restores the default level. The tone is the panel's only sound.
ZZZ	Sleep the display	Blanks the matrix. Any button wakes it; an arriving alarm wakes it once, on its rising edge. Data and history keep flowing while asleep.

SLEEP IS A CURTAIN, NOT A POWER SWITCH

A sleeping panel still receives snapshots, still accumulates history, and still paints the four-pixel alarm beacon in the top-right corner. If you walk past a dark panel with four flashing red pixels, something is down and unacknowledged.

READING THE WALL · 1 OF 2

Themes A and B

Abstract renders shape, Text renders words. Same data, two densities. VOL+ on the panel gives you the two-line version of any of these in place.

A0 · STATE LATTICE



One LED per system, whole fleet, filled by tier then name — core infrastructure first. Colour is state, brightness is tier. Answers *how much* is wrong, never *what*.

A1 · FLOW GATES



The live Ubiquiti path: APs left, hubs, switch, router, internet column, backup WAN at the right edge. Leg brightness is receive level, particles are transmit. The internet column inherits the router's state.

A2 · MQ & SIGNALS



Four lanes of travelling packets with three-cell backlog indicators — amber under half, red over. Driven by probe loss, data freshness, unknown share, and degraded share.

B0 · HEADLINE COUNTS



UP, DOWN, tier flag, DEG/MNT in words with watermark numerals. Zeroes render azure and dim; real counts take their condition colour. **TIER 0 in red means core infrastructure is down.**

B1 · THROUGHPUT



Aggregate rate as a watermark, a 24-cell utilisation bar scaled to the panel's own recent peak, and a ticker with the absolute IN / OUT / LINK figures. Bar = busy-relative-to-recent; ticker = numbers.

B2 · LOAD & LATENCY



Mean fleet load as the watermark (amber above 75%), P95 latency and loss alternating on a five-second slot, and a hot count that prints its denominator — the full panel roster, wider than B0's agent counts.

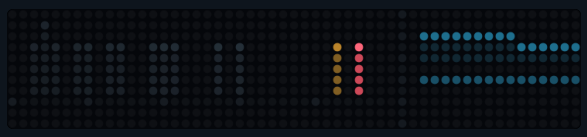
Every screen shows `NO DATA` rather than a black board before the first snapshot arrives, and prints `LINK LOST` when more than 15 s pass without a valid frame — that one is a panel fault, not a fleet fault.

READING THE WALL · 2 OF 2

Themes C and D

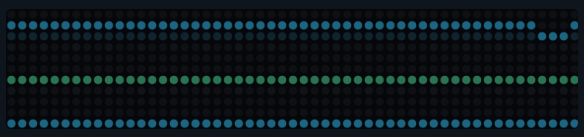
Charts render history, Instruments render texture. All history is the panel's own and is lost on reboot — a half-filled trace means a recently restarted panel, not a fleet that just woke up.

C0 · LOAD HISTOGRAM



35 buckets of where fleet load sits — left idle, right pinned, bar height is the share of systems there, bar colour is the bucket's own band. Right third: 15-sample throughput ribbon, ingress up, egress down.

C1 · TRIPLE TRACE



Three stacked 53-sample histories: throughput (azure), CPU (green), round-trip (bottom). Oldest left, now at the right edge, one column per snapshot.

C2 · POOL BARS



Up to seven pools, worst first: a one-pixel status key at the left edge, then a 46-cell mean-load bar. The key is condition, the bar is capacity — **read the key first**.

D0 · PROBE WATERFALL



Eleven worst-first systems, a new column every 0.42 s scrolling right to left — about 22 s of history. Red is a lost probe, amber is slow, azure brightness is round-trip time.

D1 · POOL FIELD



The resting face, and the boot screen. Bar length is pool size; down/degraded/unknown/maintenance stack in from the right so the worst always terminates the bar. The ticker underneath is the only place outside an alarm where a specific alert is named.

D2 · AMBIENT FIELD



A procedural field: granularity grows with fleet size, hue is mean load — azure quiet, warming past 35%, fully warm by 85%. Up to nine red sparks mark down systems; the centre readout cycles size, load, and rate.

ALARMS & THE DASHBOARD

Answering an alarm, shaping the wall

The alarm is scoped to an episode; your acknowledge is scoped to the same episode. The dashboard's Panel page is the remote control, and the panel itself has the last word on every setting.

When it sounds

1. An alarm inlay appears over the current screen, the top-right beacon flashes, and the two-tone sounds (every ~12 s while the critical is active). At night the tone auto-silences after its profile window; the light stays.
2. **Press any button on the panel to acknowledge.** The tone stops and the inlay drops for that episode. The alert itself is *not* cleared — fix the cause from the dashboard as usual.
3. If a *new* episode arms (a new critical, or the old one cleared and re-fired), the panel sounds again. That is by design.

The Panel page (dashboard.akoria.net → Panel)

CONTROL	WHAT IT DOES
Theme / Screen	Jump the wall to any screen. The live tile mirrors what the wall is showing right now, with its blurb.
Brightness · AUTO	Manual percentage, or AUTO to let the light sensor drive it. Re-enabling AUTO releases the latch a LUX button set.
Dwell 3 / 6 / 30 s	The base interval each screen's weight multiplies. 6 s is the boot default.
Rotation per screen	On/off on each screen's tile. A skipped screen is greyed and badged.
Weight ¼× – 10×	Per-screen dwell multiplier (¼, ½, 1, 2, 5, 10). Weighted dwell = base dwell × weight.
Acknowledge	Remote ack of the current episode — same scope as pressing a button on the wall.
Sleep / Wake	Explicit. Other commands deliberately do not wake a sleeping panel.

Pending vs confirmed. A command shows *pending* until the panel's own state report confirms it — the panel is the sole authority. Commands expire after 120 s if the panel never confirms; screen enable/weight changes persist on the panel across reboots and reflashes.

WHEN IT MISBEHAVES

Troubleshooting

Three verdicts the panel prints about itself, and the ninety-second checks before anyone opens a terminal. The deep version is the Operator Manual's LINK LOST ladder.

SYMPTOM	MEANING	DO THIS
NO DATA	The panel is healthy but has never received a snapshot since power-up.	Wait ~10 s after boot. If it persists, the daemon on lithium isn't reaching the API — check `solari-panel` service.
LINK LOST	More than 15 s without a valid frame. Panel fault (USB/daemon), not fleet fault.	Reseat the USB cable — the link auto-relinks on replug. Still lost: restart `solari-panel` on lithium.
STALE flag on data	Snapshots arriving, but xenon's data is old — a monitoring-side gap.	Check the dashboard; the fleet view will show the same staleness.
Wall ignores commands ~40 s after a daemon restart	The daemon's wire sequence restarts and the panel rejects frames until it catches up (adopts after ~30 s).	Nothing — it heals itself. Don't power-cycle the panel for this.
Screens look dim in daylight	Manual brightness latched low, or auto-brightness ceiling.	Re-enable AUTO from the Panel page, or LUX + on the wall.
Alarm too quiet	VOL− was pressed; there is no volume-up button.	Power-cycle the panel to restore the default level (persisted settings survive; volume resets by design).
A screen never appears	Rotation disabled or weighted down for that screen.	Panel page → that screen's tile → enable, or raise its weight.

BOOT DEFAULTS

Boots on D1 · dwell 6 s · auto-brightness on · alarm volume default · per-screen enables and weights restored from the panel's own flash (they survive reboot and reflash).

FURTHER READING

Operator Manual — every screen, pixel by pixel, plus the alarm anatomy and the full troubleshooting ladder. Operator Quickref — the panel section alongside the rest of the fleet's runbooks. Both are served from the dashboard host and print to US Letter.